

Day 01

Exploring Data with `pandas`

MSU REU Machine Learning Short Course

The Dataset — SDSS

Sloan Digital Sky Survey: 100,000 observations, 17 features, three classes.

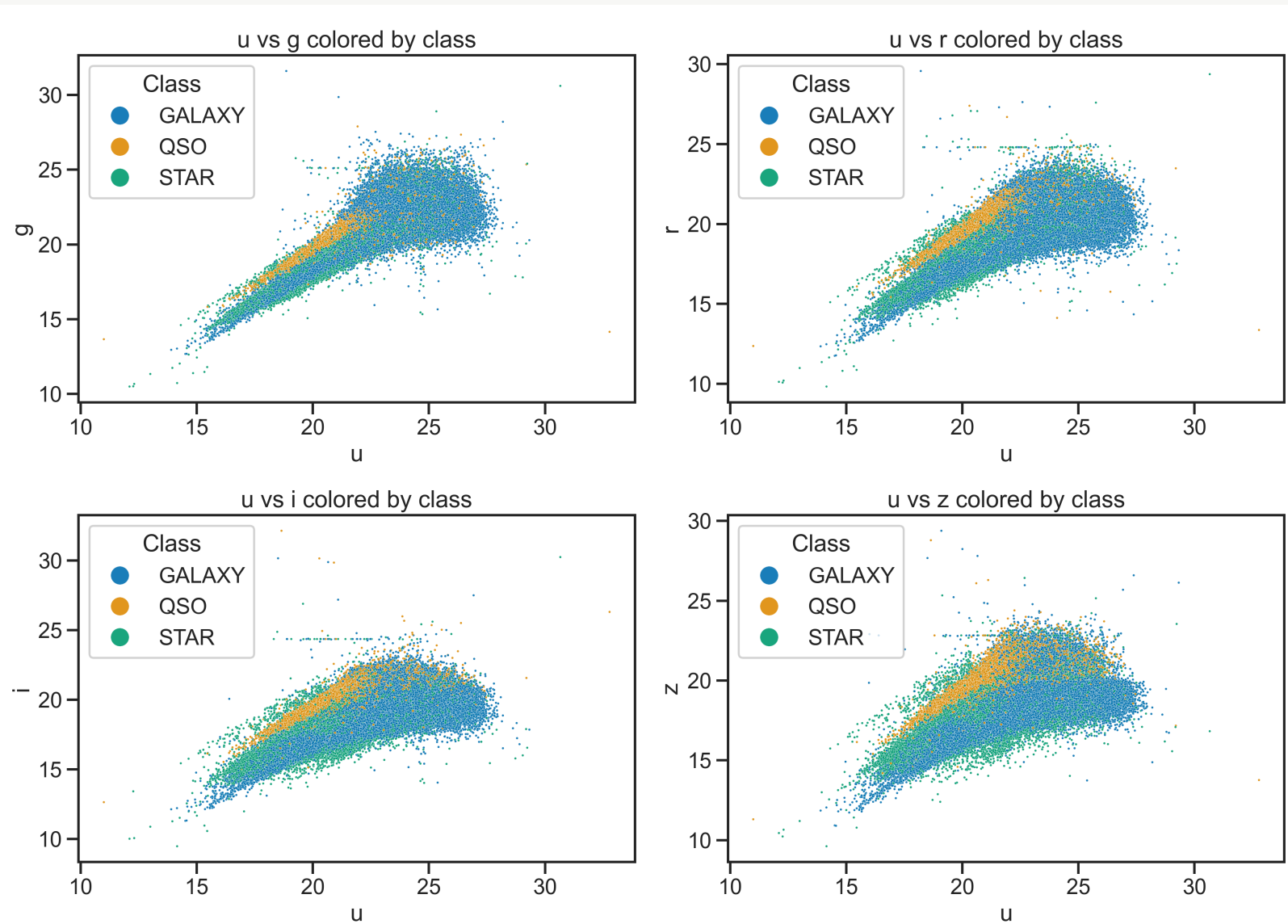
Class	What it is
STAR	A nearby star in our galaxy
GALAXY	A distant galaxy
QSO	A quasar — the bright core of a very distant galaxy

We'll use this same dataset all week.

The dataset is at `activities/data/star_classification.csv`

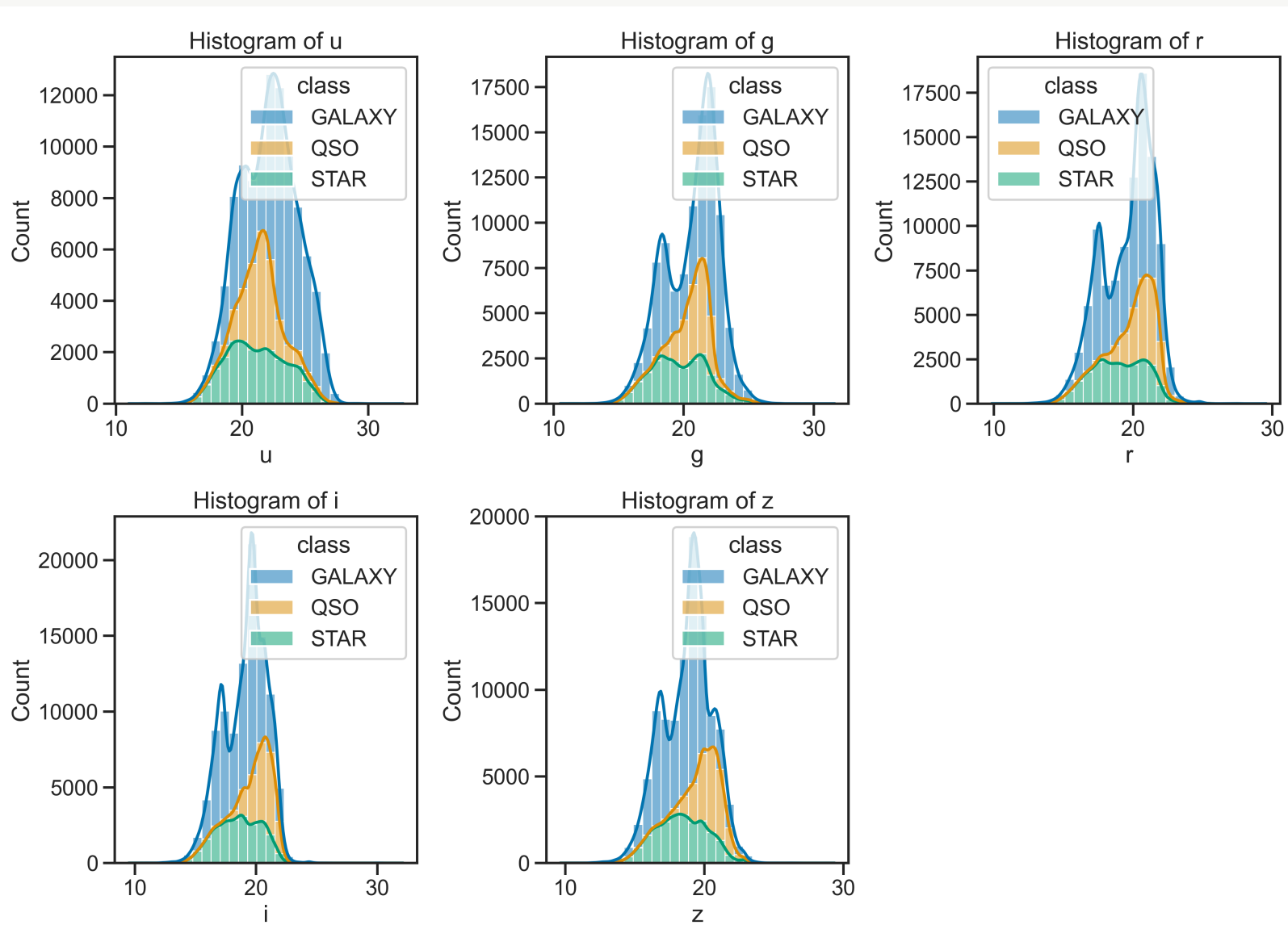
What does the data look like?

Five photometric color bands: `u g r i z` — brightness at different wavelengths.



Distributions by class

Redshift z separates classes dramatically. Color bands? Less so.



The Scientific Python Stack

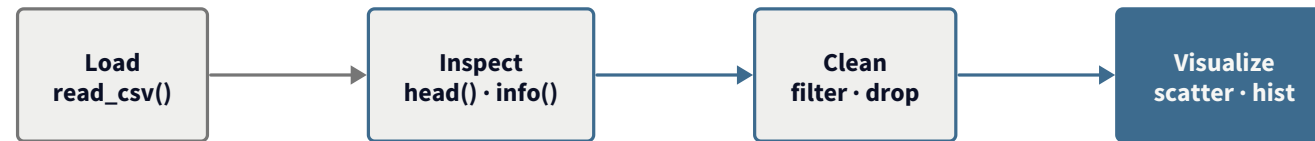
We'll use three libraries throughout the course.



- **numpy** — fast array math, the foundation everything else builds on
- **pandas** — labeled tables (like a spreadsheet, but scriptable)
- **matplotlib / seaborn** — plots

The EDA Loop

Exploratory data analysis always follows the same four steps.



You'll run this loop many times — each pass teaches you something new.

Key pandas commands

```
import pandas as pd

df = pd.read_csv("data/star_classification.csv") # load
df.head()          # first 5 rows
df.info()          # column types, nulls
df.describe()      # summary statistics

# filter: remove rows where any band is negative
df = df[df["u"] > 0]

# select columns
features = df[["u", "g", "r", "i", "z"]]
```

Today's Activity

Work through **Activity 01: Exploring Data with pandas**

1. Load and inspect the SDSS dataset
2. Clean it — remove invalid photometric values
3. Make scatter plots and histograms colored by class
4. Answer: which features separate classes best?

Before you start: [Python Review](#) if you need a refresher

Docs: [pandas](#) · [seaborn](#)